

KARTHIK PASULA

Backend Engineer | Python · FastAPI · PostgreSQL · Docker · REST APIs
+91 8790518267 | pasulakarthik03@gmail.com | github.com/Pasulakarthik | linkedin.com/in/karthik-pasula | Nandyal, India

PROFESSIONAL SUMMARY

Backend Engineer with hands-on experience designing, building, and deploying secure, scalable REST APIs using Python, FastAPI, PostgreSQL, and Docker. Independently architected a JWT + RBAC authentication system with 15+ endpoints and achieved a 35% reduction in database query response time through SQLAlchemy ORM optimisation and indexed queries. Proficient across the full backend development lifecycle — API design, relational database modelling, containerised deployment, testing with pytest, and Linux server environments. Seeking a junior backend or associate software engineer role to contribute to production-grade distributed systems.

TECHNICAL SKILLS

Languages	: Python, SQL, Bash Scripting
Backend Frameworks	: FastAPI, REST API Design, API Integration, Pydantic, Swagger, HTTP/HTTPS, JSON
Authentication & Security	: JWT (JSON Web Tokens), Role-Based Access Control (RBAC), argon2 Password Hashing, Secure Session Management, Input Validation
Databases	: PostgreSQL, SQLAlchemy ORM, RDBMS, SQL Query Optimisation, Indexing, CRUD Operations
Testing	: pytest, Unit Testing, Test-Driven Development (TDD)
DevOps & Tools	: Docker, Containerisation, Git, GitHub, Linux, Render, VS Code
Concepts	: Clean Architecture, OOP, Scalability, API Security,

PROJECTS

FastAPI Authentication API *Python · FastAPI · PostgreSQL · SQLAlchemy · JWT · Docker · Pydantic · argon2 · RBAC*
GitHub: https://github.com/Pasulakarthik/new_api | Live API: <https://new-api-k30v.onrender.com/>

- Designed and deployed a production-ready RESTful API with 15+ endpoints following clean architecture and separation-of-concerns principles, enabling scalable and maintainable backend services.
- Implemented JWT-based authentication and Role-Based Access Control (RBAC) to enforce granular access policies and secure resource authorisation across all endpoints.
- Optimised database performance by 35% via SQLAlchemy ORM query tuning and strategic indexing on PostgreSQL, reducing average query response time under load.
- Applied argon2 password hashing, secure session management, and Pydantic data validation to ensure data integrity and protect against common web vulnerabilities.
- Containerised the full application stack using Docker, enabling consistent local development and reproducible production deployments on Render.
- Wrote unit tests with pytest to validate endpoint behaviour, authentication flows, and edge-case handling.

System Monitoring Tool *Bash · Linux*
GitHub: <https://github.com/Pasulakarthik/Bash-Projects>

- Built a Bash script to monitor CPU, memory, and disk utilisation in real time with configurable alert thresholds, improving infrastructure observability.
- Implemented automated logging and threshold-breach notifications, reducing manual health-check effort by an estimated 80% and enabling proactive incident response.

Docker Deployment Automation

- Automated a complete Docker pipeline — image pull, container restart, port mapping, and environment configuration — via Bash scripts, reducing manual deployment steps from ~15 commands to a single script execution.
- Improved deployment reliability and repeatability, laying groundwork for CI/CD pipeline integration.

Log Analyzer

- Built a Bash-based log parsing tool to extract web server traffic insights, error statistics, and usage patterns, automating routine analysis that previously required manual grep pipelines.

Frontend Clones — Bigbasket & Flipkart *HTML · CSS*
Bigbasket: <https://pasulakarthik.github.io/bigbasket/bigbasket.html> | Flipkart: <https://pasulakarthik.github.io/flipkart/flipkart.html>

- Built responsive homepage clones of Flipkart and Bigbasket using HTML and CSS, recreating navigation bars, product sections, banners, and layouts with responsive design for multiple screen sizes.

EDUCATION

Bachelor of Science in Computer Science (B.Sc. CS) *April 2026*
Sri Ramakrishna Degree College (Autonomous) — Nandyal, India *CGPA: 8.00 / 10.0*

CERTIFICATIONS

Python Programming *Reliance Foundation Skilling Academy | 2025*
Communication Skills *APSSDC | 2024*